



Slide 1: Problem – Dialysis Wastes High-Quality Water

Headline: Hospital dialysis in France discards millions of liters of high-grade water that could safely replace non-potable uses.

- France has about 51,700 dialysis patients; over half of all renal replacement patients are on dialysis, with 77% in non-autonomous in-center/UDM units, i.e. mainly hospital or satellite facilities.^{[1] [2] [3] [4] [5]}
- A typical haemodialysis session uses around 400–500 L of mains water, of which 50–70% is rejected by the reverse-osmosis (RO) unit as clean greywater that is normally sent to drain.^{[6] [7] [8] [9]}
- Large, multi-station dialysis centers in university hospitals (32 CHU/CHRU) and big private clinics are few in number but treat a large fraction of patients, making them the natural first adopters of water-reuse infrastructure.^{[10] [11] [12]}
- Case studies in European hospitals show RO-reject recovery systems can save 3,000–5,000 m³ of water per year per unit with payback times of roughly 1–3.5 years, using the water for toilet flushing, laundry, or horticulture.^{[13] [9] [14]}
- France's drinking water production has an estimated carbon footprint of roughly 0.23–0.27 kgCO₂e per m³, so reusing RO-reject instead of new potable water avoids both water abstraction and associated emissions.^[15]

Slide 2: France Market Size and Segmentation

Key facts (system size and concentration)

- At end-2024, France had 2,965 health establishments: 1,330 public hospitals, 655 private non-profit establishments and 980 private for-profit clinics.^[16]
- In 2022, 93,084 patients were on renal replacement therapy; 55.5% were on dialysis (≈51,700 patients), the rest living with a functioning graft.^{[4] [5]}
- Modality split from the REIN registry/France Rein shows 51.7% of dialysis patients in in-center hospital HD, 26% in medicalized dialysis units (UDM), and the remainder in self-dialysis, home HD (~1.5%), and peritoneal dialysis.^{[2] [3] [1]}

Where activity is concentrated (for water-reuse potential)

- Dialysis in France is delivered in about 1,100–1,200 units; a network analysis identified 1,166 dialysis units in 2018, organized into ~160 regional “communities” of care.^[12]
- Large in-center HD and UDM facilities (often CHU/large CH plus big private dialysis centers) run many stations and multiple shifts, so they deliver far more sessions per site

than small satellite/self-dialysis units.^{[11] [2] [12]}

- Given that non-autonomous in-center/UDM accounts for ~77% of dialysis patients, most water-reuse potential is concentrated in a few hundred large or medium-sized hospital-based units rather than in dispersed home or small self-dialysis facilities.^{[3] [1] [2]}

Slide-ready table – market and first adopters

Hospital / site type	Role in dialysis market (approximate)	First-adopter potential for RO-reject reuse
CHU/CHRU (32 university hospitals)	Large nephrology departments, multi-station HD units, high patient throughput. ^{[10] [11]}	Very high: concentrated sessions, technical teams, capex capacity.
Large general public hospitals (CH) with HD/UDM	Many UDM and in-center units integrated in regional networks. ^{[1] [2] [12]}	High: enough volume for ROI; can pool investment at GHT level.
Large private dialysis centers/clinics	Significant share of in-center and UDM activity under a few operators. ^{[2] [17]}	High: clear P&L incentives and standardized technical platforms.
Medium hospitals with small HD rooms/UDM	Lower station counts, still non-autonomous dialysis for local catchment. ^{[1] [12]}	Selective: feasible if integrated into broader hospital grey-water plan.
Small hospitals, self-dialysis units, home HD	Fragmented, limited patients per site; some without full technical services. ^{[2] [1]}	Low: capex per saved m ³ too high; complexity outweighs benefits.

Slide 3: Technology Options – Ranked and Scored

Context (why water reuse is realistic but bounded)

- Haemodialysis uses treated water to prepare dialysate; RO systems reject 50–70% of incoming water, which never contacts blood or dialysate and is therefore relatively clean (low microbial risk, but not potable-grade).^{[18] [7] [6]}
- International experience shows that capturing RO-reject and routing it to grey-water uses (toilets, irrigation, some cleaning/laundry) is technically straightforward when separated from potable circuits and designed with estates/engineering teams.^{[19] [9] [14] [13]}
- More ambitious concepts—reuse of spent dialysate or heat recovery—are discussed in circular-economy nephrology papers but have limited real-world deployment and need further validation.^{[20] [14] [21]}

Ranking and simple scoring (1–5; 5 = best)

Scores are expert judgement based on current literature and case studies; they indicate relative positioning, not precise metrics.

1. Capture and reuse of RO-reject water for toilet flushing and irrigation (near-term)

- How it works: RO-reject is piped from dialysis water plant to a storage tank, then pumped to a separate grey-water network feeding toilet cisterns and/or landscape irrigation; bypass to drain during disinfection or tank overflow.^{[9] [14] [13]}

- Maturity: Proven in multiple hospitals (UK, Ireland, Malaysia) for 10+ years, with documented volumes, costs and payback.^{[14] [9]}
- Safety: Low infection risk because water never re-enters clinical circuits or potable water; main concerns are cross-connection control and legionella management in storage/distribution.^{[20] [9] [14]}
- Fit: Best for large and medium hospitals with big dialysis units and sizable toilet/irrigation demand.
- Scores: Impact 5 / Ease 4 / Safety 4 / Cost 4 / Time-to-pilot 5.

2. RO-reject reuse for cleaning, technical uses, and selected laundry processes (near- to mid-term)

- How it works: As above, but water is also used for floor cleaning, certain washer-disinfectors, cooling towers, or pre-wash cycles where hardness and conductivity are acceptable.^{[19] [14] [20]}
- Maturity: Some case reports (e.g. reject feeding laundries, hot-water systems) but less standardized; needs site-specific engineering and compatibility checks with equipment.^{[14] [19]}
- Safety: Requires risk assessment for biofilm/scale; must not be used where drinking-water standards are required or where aerosols pose infection risk without adequate treatment.^{[20] [14]}
- Fit: Large hospitals with technical services and multiple non-clinical water demands; less suitable for small sites.
- Scores: Impact 4 / Ease 3 / Safety 3 / Cost 3 / Time-to-pilot 3.

3. Optimizing RO efficiency and dialysis water demand (complementary, near-term)

- How it works: Use higher-efficiency RO units, adjust recovery rate, minimize rinse cycles, and optimize dialysate flow; larger centers already show lower water use per session when plants are well-chosen and run continuously.^{[18] [11] [6]}
- Maturity: Well described in environmental nephrology reviews; many manufacturers already offer higher-recovery systems and smart controls.^{[7] [11] [6]}
- Safety: Must not compromise water quality; changes are usually within manufacturer specifications and under existing standards.^{[11] [18]}
- Fit: All hospital sizes; strongest gains in large centers.
- Scores: Impact 3 / Ease 4 / Safety 4 / Cost 4 / Time-to-pilot 4.

4. Treatment and reuse of mixed haemodialysis wastewater (spent dialysate + other streams) for irrigation or technical uses (mid-term)

- How it works: Collect total haemodialysis wastewater, treat via coagulation, sedimentation and ultrafiltration, then reuse for irrigation or other non-potable uses; RO-reject alone can also be used where salinity permits.^{[22] [19]}
- Evidence: An Iranian hospital study showed CFS+UF treatment could meet reuse criteria for several parameters, but high salinity and organic load limited direct irrigation without focusing specifically on RO-reject.^{[22] [19]}

- Safety/complexity: Higher; involves handling potentially contaminated effluent, more complex treatment trains, and stricter monitoring; likely too complex for initial pilots in France.
- Fit: Larger centers with ambitious sustainability programs; probably not first wave.
- Scores: Impact 4 / Ease 2 / Safety 2 / Cost 2 / Time-to-pilot 2.

5. Reuse of spent dialysate and thermal energy recovery (long-term)

- How it works: Concepts include purifying spent dialysate for toilet flushing or other uses and installing heat exchangers to recover thermal energy from warm effluent. [\[21\]](#) [\[14\]](#) [\[20\]](#)
- Evidence: Mostly conceptual and small-scale; the 2024 circular-economy point-of-view paper lists these as future opportunities, not standard practice; no large hospital-wide deployments reported. [\[21\]](#) [\[14\]](#) [\[20\]](#)
- Safety/complexity: High, as dialysate effluent contains blood traces, solutes and potentially pathogens; any reuse needs robust treatment, validation and regulatory approval.
- Fit: Research centers or flagship university hospitals; not ready for mainstream deployment.
- Scores: Impact 5 / Ease 1 / Safety 1 / Cost 1 / Time-to-pilot 1.

Slide 4: Evidence – Impact, Savings, and Uncertainty

Water consumption and reject volumes (well-supported)

- Multiple technical and clinical sources converge on ~400–500 L of mains water per standard 4-hour HD session, with RO systems rejecting roughly 50–70% of feed water; that implies 200–350 L of RO-reject per session. [\[8\]](#) [\[6\]](#) [\[7\]](#) [\[9\]](#) [\[18\]](#)
- The 2025 Clinical Kidney Journal environmental impact study of 20 HD centers reported average water consumption of about 600 ± 200 L per session, with larger centers using less water per session than smaller ones. [\[23\]](#) [\[11\]](#)
- A global review estimated total HD water use at 265 million m³/year for one million hemodialyzed patients, with about two-thirds as RO-reject, consistent with ~0.5 m³ of total water per session. [\[6\]](#) [\[18\]](#)

Case studies of RO-reject reuse (strong for feasibility and economics, mostly outside France)

- The Canterbury dialysis unit (UK) installed a system in 1999 that recovers about 800 L/h of RO-reject, piping it to a grey-water tank to flush hospital toilets; capital cost was ~£15,000, with annual water/sewer savings of ~£7,500 and payback in ~2 years. [\[13\]](#) [\[9\]](#) [\[14\]](#)
- Other hospitals (e.g. Countess of Chester, Lister, Maidstone, Sultan Abdul Halim in Malaysia) report recovered volumes of 1,460–5,240 m³/year, investment costs around 6,000–14,000 (USD equivalent) and payback between ~12 and 42 months. [\[14\]](#)
- A case study from Saveh hospital in Iran showed that focusing on RO-reject alone (2,300 L/d) could irrigate ~287.5 m² of hospital green space, whereas fully treated mixed

haemodialysis wastewater remained too saline for direct irrigation, reinforcing the focus on RO-reject as the most practical fraction.^{[19] [22]}

Order-of-magnitude impact for a French large hospital (reasoned estimate)

Assumptions clearly stated:

- 100 chronic in-center/UDM patients in a large hospital; 3 sessions/week; 156 sessions/patient/year → 15,600 sessions/year.
- 270 L RO-reject/session as a conservative mid-point within the 200–350 L range.^{[7] [8] [9] [6]}
- **Water savings:** $15,600 \times 0.27 \text{ m}^3 \approx 4,200 \text{ m}^3/\text{year}$ of potable water avoided if all RO-reject is reused for toilets/irrigation (assumption based on linear scaling from international case studies).^{[9] [14]}
- **CO₂e savings:** Using 0.23–0.27 kgCO₂e/m³ for water supply in France, this corresponds to roughly 1.0–1.1 tCO₂e avoided per hospital-year, excluding wastewater treatment emissions (estimate extrapolated from drinking-water CF data).^[15]
- **Financial savings:** If combined water and sewer tariffs are on the order of a few €/m³ (typical of European urban tariffs; assumption, no France-specific figure used), 4,200 m³/year implies low five-figure euro savings per year; UK sites realized ~£7,500/year savings at similar scales.^{[9] [14]}

Scaling to the French system (illustrative, not precise)

- If 100–150 large and medium hospitals implemented RO-reject reuse achieving 3,000–5,000 m³/year each (range of published case studies), total savings would be 300,000–750,000 m³/year, corresponding to roughly 70–200 tCO₂e/year from water supply alone.^{[15] [9] [14]}
- These values are modest at national scale but material at hospital level and symbolically important for “green nephrology”; they also come with no clinical trade-off when kept strictly in non-clinical circuits.^{[11] [20] [14]}

Confidence levels (for your pitch)

- **High confidence:** Dialysis water use per session; fraction of RO-reject; total dialysis population and modality split; technical feasibility and payback (1–3.5 years) of RO-reject reuse for toilets/irrigation in large centers; relative environmental benefit of focusing on in-center/UDM units.^{[5] [2] [18] [6] [11] [9] [14]}
- **Moderate confidence:** Extrapolated per-hospital French water/CO₂ savings (depends on exact patient load, water tariffs, and RO settings); assumption that 100–150 hospitals represent most of the economic potential.
- **Low confidence / exploratory:** Large-scale reuse of spent dialysate and thermal energy in French hospitals; reuse of fully treated haemodialysis wastewater for sensitive uses beyond irrigation/technical processes.^{[22] [20] [19] [14]}

Slide 5: Roadmap – From Pilot to Scale

Table – staged roadmap

Horizon	Most realistic actions	Expected impact (per pilot hospital)	Main blockers	What must be proven next
0–6 months	Select 1–3 large hospitals (CHU/large private centers) with big HD units; perform water audit and mapping of existing RO plants and toilet/irrigation circuits; co-design RO-reject capture system with technical services; secure infection-control and regulatory green light. [11] [9] [14]	Detailed baseline of water use per session and per hospital; engineered design for ~3,000–5,000 m ³ /year potential reuse; clear capex/opex budget and projected 2–6 year payback based on local tariffs (estimate). [9] [14]	Capital budgeting cycles; competing priorities for estates teams; limited local experience with grey-water systems; need for clear infection-control guidance.	Practical design patterns for French hospitals; validated risk-assessment and maintenance protocols; internal buy-in from nephrology, estates, and infection-control.
~3 years	Implement RO-reject reuse in 10–20 large/medium hospitals; link projects to regional GHT or private group decarbonization plans; integrate procurement templates and standard technical specs; begin exploring added uses (cleaning, laundries) where safe and economical. [11] [20] [14]	Systematic savings of perhaps 40,000–100,000 m ³ /year across early adopters; avoided emissions on the order of tens of tCO ₂ e/year; learning curve on capex, reliability, and maintenance; visible “green nephrology” results for public reporting (estimates). [15] [9] [14]	Heterogeneity of existing buildings; fragmentation of decision-makers; limited national incentives focused specifically on dialysis water; need for robust monitoring data to convince clinicians and regulators.	Comparative performance data across sites; validated business case under French tariffs; evidence that patient care and infection-control are unaffected or improved.

Horizon	Most realistic actions	Expected impact (per pilot hospital)	Main blockers	What must be proven next
5–7 years	Integrate RO-reject reuse into standard design/renovation of dialysis units and hospital water systems; extend to more advanced options (e.g. partial reuse of dialysate effluent, heat recovery) in a few research-oriented centers; embed water-reuse KPIs in national quality/sustainability frameworks. [20] [14] [21]	Structural reduction of dialysis-related water use in much of the in-center/UDM sector; cumulative savings of hundreds of thousands of m ³ and hundreds of tCO ₂ e; replication of best practice beyond nephrology (e.g. sterilization, imaging cooling). [11] [20] [14] [15]	Regulatory and standards adaptation for more complex reuse (dialysate), technology cost, and technical complexity; ensuring smaller sites are not left behind in terms of support and templates.	Safe and economically viable models for dialysate/thermal energy reuse; national/European guidance recognizing dialysis water reuse as good practice; tools to support smaller hospitals (shared engineering/procurement).

Slide 6: Business Model and Equity

Who pays, who benefits, who operates?

- **Buyer / decision-maker:** Hospital management (public CHU/CH and private dialysis groups), typically via estates/engineering, within regional hospital groups (GHT) or private chains; they control capex budgets and sustainability objectives. [\[24\]](#) [\[16\]](#) [\[11\]](#)
- **User / operator:** Dialysis unit staff (as primary water producers) and hospital technical services who maintain RO plants, storage tanks, pumps and grey-water networks. [\[11\]](#) [\[9\]](#) [\[14\]](#)
- **Beneficiary:**
 - Direct: Hospital (reduced water and sewer bills, better environmental reporting), dialysis services (green nephrology reputation), and hospital groups aiming at decarbonization targets. [\[20\]](#) [\[14\]](#) [\[11\]](#)
 - Indirect: Local water utilities and communities (reduced demand on potable water and wastewater treatment), and patients who may value sustainable care. [\[18\]](#) [\[6\]](#) [\[20\]](#)
- **Partner:**
 - Water/energy utilities and engineering firms that design/install water-reuse systems.
 - Medical device/RO manufacturers offering higher-recovery systems and integration with sustainability modules. [\[7\]](#) [\[11\]](#)
 - Regional health authorities (ARS) and Agence de la transition écologique for grants or inclusion in climate plans. [\[11\]](#) [\[20\]](#)

- **Maintainer:** Hospital estates/biomedical engineering teams, possibly with maintenance contracts to specialized water-treatment providers.^{[19] [9] [14]}

Economic value proposition

- For large centers, RO-reject reuse is essentially a **utility-savings project**: capex for tanks, piping, pumps and controls (tens of thousands of euros) repaid over a few years via avoided potable water and sewer costs, as suggested by international payback data (1–3.5 years) and French water-carbon intensities.^{[15] [9] [14]}
- Additional value comes from regulatory and reputational benefits: alignment with French and EU climate policies, contributions to hospital-level sustainability KPIs, and demonstration of responsible resource use in a highly resource-intensive therapy.^{[14] [20] [11]}

Procurement and implementation logic

- Start with **performance-based procurement**: suppliers commit to delivering a defined amount of recoverable water (m³/year) and providing remote monitoring of flows and quality, while hospitals retain ownership of savings; options for shared-savings contracts can be explored but require legal work.
- Use **framework contracts** at GHT or private-group level to standardize technical specifications, risk assessments, and maintenance models, thus lowering transaction costs for each hospital.^{[14] [11]}

Equity and access implications

Separated into four dimensions as requested:

- **Environmental efficiency:** Focusing on large hospitals maximizes liters saved per euro and per engineering hour; small sites are less efficient targets, which is environmentally logical but risks a perception gap.^{[11] [14]}
- **Clinical feasibility:** Because water reuse is strictly non-clinical (toilets, irrigation, cleaning), there is no direct difference in clinical access or modalities between patients; however, pilots must demonstrate that projects never interfere with water quality for dialysis itself.^{[18] [20] [11]}
- **Social fairness (Paris vs regions, rich vs poor hospitals):** Early pilots will naturally be in better-resourced CHU and big private centers (often urban/Paris-centric), potentially widening perceived inequalities in “green” infrastructure; this can be mitigated by making all technical designs, protocols and data open and reusable for regional hospitals, and by including at least one non-Paris regional CHU among first pilots.^{[10] [12] [11]}
- **Geographic equity and feasibility for smaller hospitals:** Smaller hospitals and rural UDMs may lack capex and engineering capacity; to avoid exclusion, regional authorities or GHT could co-finance “template” projects and offer regional engineering support or shared maintenance contracts, enabling scaled-down implementations where volume justifies it.^{[12] [11]}

Design principles to improve, not worsen, access

- Make all tools (design drawings, risk assessments, monitoring dashboards) openly shareable across hospitals to reduce knowledge asymmetry.
- Tie funding to inclusion criteria that ensure at least some peripheral/regional hospitals participate in the second wave, once the model is proven in flagship centers.
- Explicitly track staff workload and maintenance burden in pilots to show that solutions are manageable for smaller teams and to refine designs where they are not. ^{[20] [14] [11]}

Slide 7: Final Pitch Message (20–30 seconds)

“Dialysis keeps people alive, but every session in France wastes hundreds of liters of high-quality water straight to the drain. By capturing this RO-reject and reusing it for toilets and irrigation in large hospital dialysis units, we can save thousands of cubic meters of water per hospital each year, cut CO₂ emissions, and reduce utility costs—without touching the clinical circuit. Our proposal is to pilot this in a small group of CHU and large dialysis centers, generate robust operational and economic data, then scale a standard, equity-minded model across France’s hospital networks.” ^{[2] [8] [10] [6] [9] [15] [14] [11]}



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